

Ch 1 - Functions

Terminology:

Function – a mapping where each input value gives **only one output value**

Domain – the set of input values for a function

Example: $f(x) = x + 2$ for $x > 0$; the domain is $x > 0$

Range (image set) – the set of output values for a function

Example: $f(x) = x + 2$ for $x > 0$; the range is $f(x) > 2$

One-one function – **one** input value gives **one** output value

Example: $f(x) = x + 3$

Many-one function – **many** input values give **one** output value

Example: $f(x) = x^2$

Inverse function – a function that reverses the original function

Example: $f(x) = x + 2$; $f^{-1}(x) = x - 2$

Composition of functions – a function formed from multiple functions being combined

Example:

$$f(x) = x + 2; g(x) = x^2$$

$$fg(x) = x^2 + 2$$

Important: A given function is a function only when an input value gives **only one output value**

Notation:

$$f(x) = 2e^x$$

$f: x \mapsto \lg x$, for $x > 0$ is the same as $f(x) = \lg x$, for $x > 0$

The inverse of $f(x)$ is $f^{-1}(x)$

$$fg(x) = f(g(x))$$

$$f^2(x) = ff(x) = f(f(x))$$

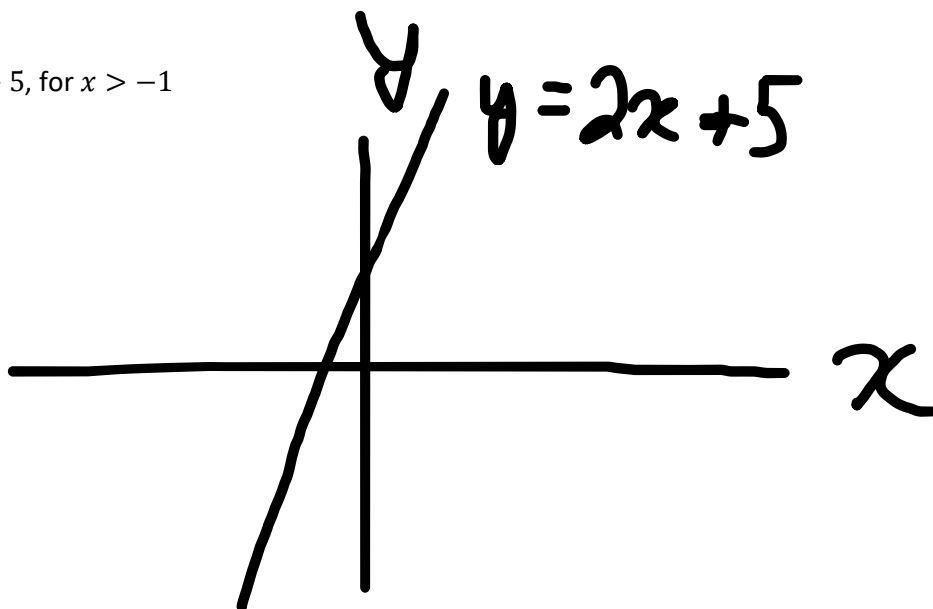
Domain and Range:

To find the range of a function, given the function and its domain:

1. Sketch the graph to obtain the function's shape
2. Find the value of the function at the highest/lowest point in the domain
3. Determine which direction the function extends to
4. State the range

Example:

$$f(x) = 2x + 5, \text{ for } x > -1$$



$$f(-1) = 2(-1) + 5 = -2 + 5 = 3$$

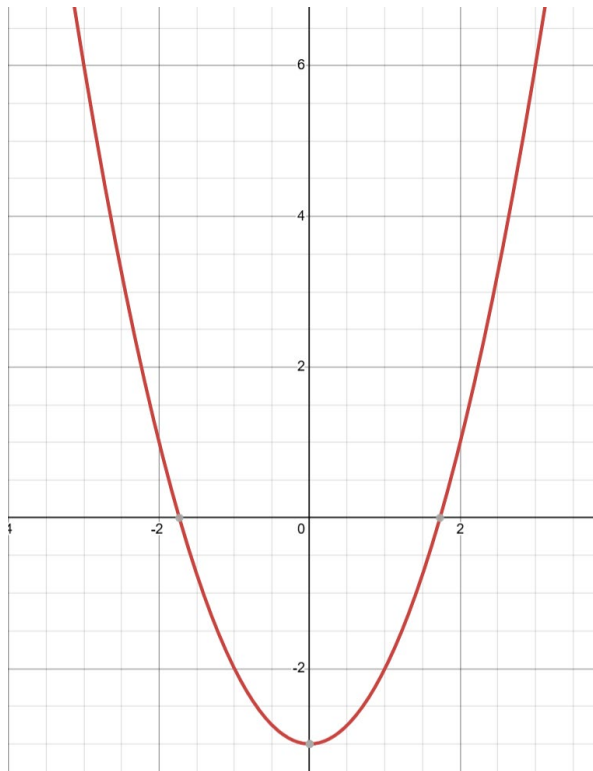
Range: $f(x) > 3$

Modulus functions:

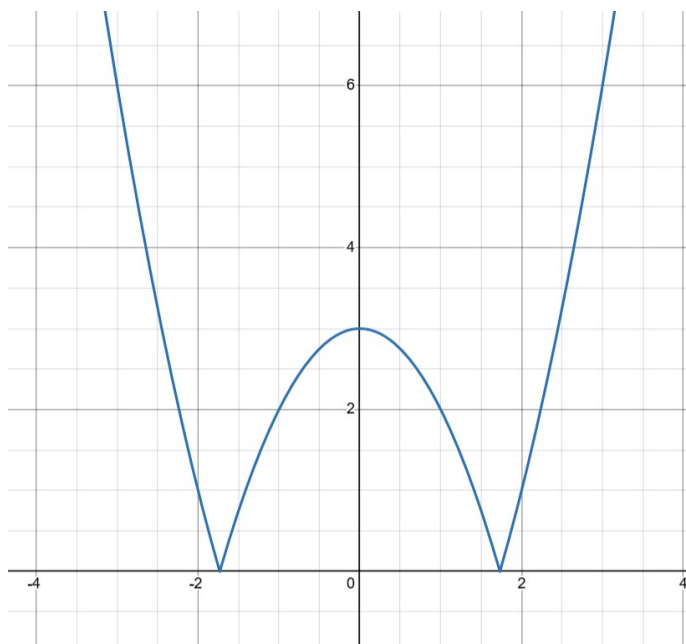
Given the function $y = f(x)$, $y = |f(x)|$ is such that any points where $f(x) < 0$ are reflected in the x -axis

Example:

$$f(x) = x^2 - 3$$



$$|f(x)| = |x^2 - 3|$$



Inverse functions:

To find the inverse of a function:

1. Write the function in terms of x and y
2. Swap the x and y
3. Rearrange the equation in terms of y

Example:

$$f(x) = 2x + 5$$

$$y = 2x + 5$$

$$x = 2y + 5$$

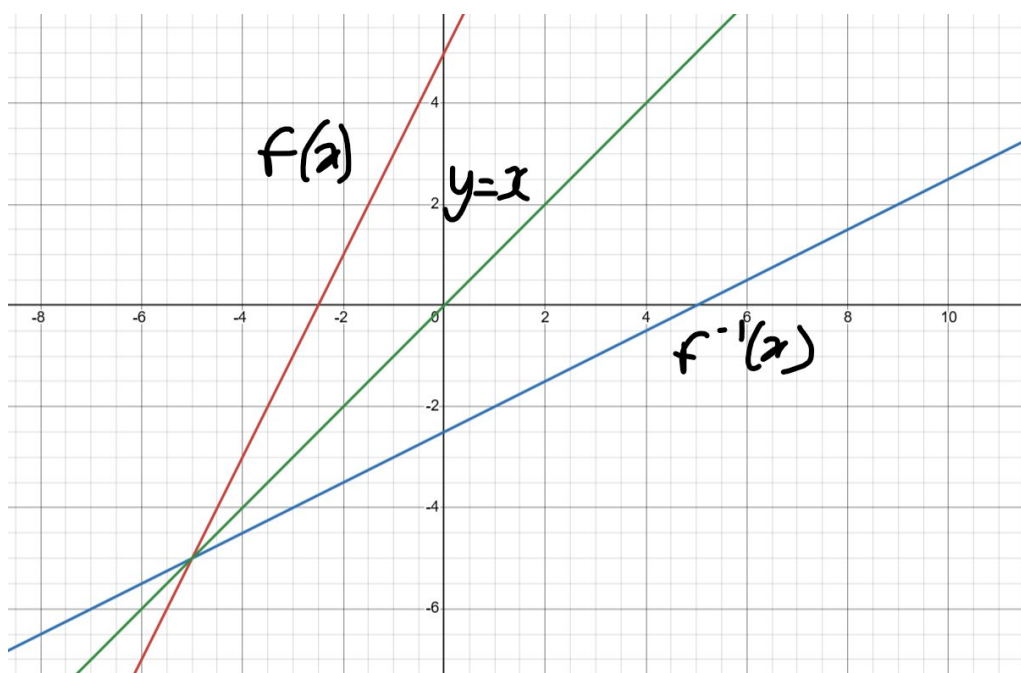
$$y = \frac{x - 5}{2}$$

$$f^{-1}(x) = \frac{x - 5}{2}$$

The domain of $f(x)$ = the range of $f^{-1}(x)$

The range of $f(x)$ = the domain of $f^{-1}(x)$

A function and its inverse are reflections of each other in the line $y = x$



Important: A given function has an inverse only when it is a one-one function (if a function is not one-one, it does not have an inverse)

Example:

$f(x) = x^2$ is a many-one function

$f^{-1}(x) = \pm\sqrt{x}$ is a one-many mapping, it is **not** a function

For $f(x) = x^2$ to have an inverse, we need to restrict its domain such that it becomes a one-one function

Example:

$f(x) = x^2$, for $x \geq 0$

$f^{-1}(x) = \sqrt{x}$, for $x \geq 0$

Composite Functions:

Example:

$f(x) = 2x + 5$

$g(x) = x^2 + 6x + 9$

$fg(x) = 2(x^2 + 6x + 9) + 5 = 2x^2 + 12x + 23$

$gf(x) = (2x + 5)^2 + 6(2x + 5) + 9 = 4x^2 + 32x + 64$

The order of the functions is important, $fg(x) \neq gf(x)$

The domain of $gf(x) \subseteq$ the domain of $f(x)$

The range of $gf(x) \subseteq$ the range of $g(x)$

Example:

$$f(x) = 2x + 5, \text{ for } x \geq 0$$

$$g(x) = x^2 + 6x + 9, \text{ for } x \geq -3$$

$$fg(x) = 2x^2 + 12x + 23, \text{ for } x \geq -3$$

Note: If you have any questions, watch the video and then join the Discord server to ask them